

04_evaluation_results

December 2, 2025

0.1 1. Project Path Configuration.

Add the project root directory to Python's search path (`sys.path`) to ensure that local packages under the `src/` folder (such as `dataset.py`, `transforms.py`, and `loaders.py`)

can be imported correctly from within the `notebooks/` directory. ##### Rationale: - Keep imports consistent across all notebooks (`01_data_eda` → `04_evaluation_results`).

- Enable reuse of modular code without duplication.
- Maintain reproducibility and a clean experiment structure.

```
[2]: # Add project root to Python path so `src` can be imported.
import sys
from pathlib import Path

PROJECT_ROOT = Path("..").resolve() # One level up from /notebooks/
if str(PROJECT_ROOT) not in sys.path:
    sys.path.append(str(PROJECT_ROOT))

print(f'Project root added to path: {PROJECT_ROOT}')
print("Python path updated to include /src\n")
```

```
Project root added to path: /lab/px/msc-sept-24-ft-cohort-capstone-chrisanich
Python path updated to include /src
```

0.2 2. Library Imports and Environment Setup.

Import all essential libraries required for evaluation, including NumPy, Pandas, Matplotlib, PyTorch, and scikit-learn.

Define the main project directories for data, experiments, and reports, ensuring that all output folders (figures and tables)

exist before execution. Detect the available computation device (GPU or CPU) to guarantee consistent evaluation behaviour

across different hardware environments. ##### Rationale: - Load the core Python and deep learning libraries needed for model evaluation and visualisation.

- Maintain directory structure consistency across notebooks to simplify reproducibility.
- Automatically create missing output folders to prevent runtime errors when saving results.
- Select the available hardware device automatically to ensure portability between systems.

```
[3]: # Standard library imports.
import sys
from pathlib import Path
import json, csv
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# PyTorch and torchvision imports.
import torch
import torch.nn as nn
from torchvision import models

# Scikit-learn metrics for evaluation.
from sklearn.metrics import (
    accuracy_score,
    precision_recall_fscore_support,
    roc_auc_score,
    confusion_matrix,
    ConfusionMatrixDisplay,
)

# Define directory constants (consistent with 01/02/03 notebooks).
DATA_DIR      = PROJECT_ROOT / "data"
EXPERIMENTS_DIR = PROJECT_ROOT / "experiments"
POOLS_DIR      = EXPERIMENTS_DIR / "pools"
CHECKPOINTS_DIR = EXPERIMENTS_DIR / "checkpoints"
RESULTS_DIR    = EXPERIMENTS_DIR / "results"
REPORTS_DIR    = PROJECT_ROOT / "reports"
FIG_DIR        = REPORTS_DIR / "figures"
TABLE_DIR      = REPORTS_DIR / "tables"

# Ensure output directories exist (create if needed).
for d in [FIG_DIR, TABLE_DIR]:
    d.mkdir(parents=True, exist_ok=True)

# Device selection.
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
print(f"Evaluation device: {DEVICE.upper()}")
```

Evaluation device: CUDA

0.3 3. Load Preprocessing Configuration.

Load the preprocessing configuration file (`preprocessing_config.json`) generated during the data exploration stage.

Extract the image size, class mappings, and normalisation statistics for both ImageNet and CXR domains.

Verify that the file exists before continuing, as it defines parameters needed for data loading, model reconstruction, and consistent evaluation across experiments. ##### Rationale: - Ensure image size, labels, and normalisation match the training setup.

- Maintain compatibility and reproducibility across notebooks.
- Prevent evaluation errors from missing or mismatched preprocessing parameters.

```
[4]: # Load Preprocessing Config (classes and stats).
cfg_path = POOLS_DIR / "preprocessing_config.json"
if not cfg_path.exists():
    raise FileNotFoundError("Missing preprocessing_config.json (run 01_data_ed.
    ↪ipynb).")

cfg = json.loads(cfg_path.read_text())

IMG_SIZE      = int(cfg["img_size"])
CLASS_TO_INDEX = {k: int(v) for k, v in cfg["class_to_index"].items()}
INDEX_TO_CLASS = {v: k for k, v in CLASS_TO_INDEX.items()}
IMAGENET_MEAN  = cfg["imagenet_mean"]
IMAGENET_STD   = cfg["imagenet_std"]
CXR_MEAN       = cfg["cxr_mean"]
CXR_STD        = cfg["cxr_std"]

print("Config loaded:")
print("- IMG_SIZE:", IMG_SIZE)
print("- Classes:", CLASS_TO_INDEX)
```

```
Config loaded:
- IMG_SIZE: 256
- Classes: {'Pneumonia': 0, 'TB': 1, 'Normal': 2}
```

0.4 4. Build Stratified Test Split

Create a fixed 10% test split from the unified dataset using stratified sampling to preserve class balance.

Load the combined source pool and separate a reproducible subset for testing based on class labels. Save the resulting test split to disk to ensure consistency across evaluation runs and model comparisons. ##### Rationale: - Maintain proportional representation of all classes in the test set.

- Ensure reproducibility by using a fixed random seed.
- Prevent data leakage between training, validation, and test subsets.
- Store the test split for consistent evaluation across models and experiments.

```
[5]: # Load the fixed 10% test split created in 01_data_ed.ipynb
# This ensures the test set is completely independent from training.

import pandas as pd
from torch.utils.data import DataLoader
from src.dataset import CXRThreeClassDataset
from src.transforms import build_shared_transforms
```

```

# Load the precomputed test split
SPLITS_DIR = POOLS_DIR
test_df = pd.read_csv(SPLITS_DIR / "test_split.csv")

print(f"Loaded fixed test split with {len(test_df):,} images.")
print(test_df["Class"].value_counts().to_string())

# Build shared eval transforms (ImageNet by default; overridden per checkpoint,
  ↪ later)
_, eval_transforms_base = build_shared_transforms(
    img_size=IMG_SIZE,
    mean=IMAGENET_MEAN,
    std=IMAGENET_STD,
)

# Build the test dataset once (we will rebuild transforms later per checkpoint)
test_dataset = CXRThreeClassDataset(
    items=test_df,
    class_to_index=CLASS_TO_INDEX,
    transforms=eval_transforms_base,
    chexpert_root=DATA_DIR / "chexpert",
    tbx_root=DATA_DIR / "tbx11k",
)

test_loader = DataLoader(
    test_dataset,
    batch_size=64,
    shuffle=False,
    num_workers=4,
    pin_memory=True,
)

print("Test DataLoader ready.")

```

```

Loaded fixed test split with 1,353 images.
Class
Pneumonia      468
TB              467
Normal         418
Test DataLoader ready.

```

0.5 5. Model Builder

Rebuild each model architecture exactly as defined during training to ensure weight compatibility across checkpoints.

Instantiate models without pretrained weights and replace their classifier heads to match the number of output classes.

Use the same configuration as in `03_model_training.ipynb` to guarantee consistent layer structure during evaluation. ##### Rationale: - Maintain architectural consistency between training and evaluation stages. - Prevent shape mismatches when loading trained weights from checkpoints. - Allow flexible evaluation of different architectures under the same experimental setup. - Enable fair performance comparison between pretrained (transfer learning) and custom CNN models.

```
[6]: # 4. Model Builder.
# Rebuilds model architectures exactly as defined in 03_model_training.ipynb.
# Used during evaluation to ensure weight compatibility across checkpoints.
# Each model is instantiated with no pretrained weights and its classifier head
# adjusted to match the number of output classes.

def build_model(model_name: str, num_classes: int, img_size: int = IMG_SIZE) -> nn.Module:
    """
    Create an empty model with the same structure used during training.
    Args:
        model_name (str): Architecture identifier ('mobilenet_v2',
        'efficientnet_b0', etc.)
        num_classes (int): Number of output classes for the final layer.
        img_size (int): Input image size (default: from preprocessing config).
    Returns:
        torch.nn.Module: Reconstructed model ready to load weights.
    """
    if model_name == "mobilenet_v2":
        m = models.mobilenet_v2(weights=None)
        m.classifier[1] = nn.Linear(m.last_channel, num_classes)
        arch_name = "MobileNetV2"
    elif model_name == "efficientnet_b0":
        m = models.efficientnet_b0(weights=None)
        m.classifier[1] = nn.Linear(m.classifier[1].in_features, num_classes)
        arch_name = "EfficientNet-B0"
    elif model_name == "resnet50":
        m = models.resnet50(weights=None)
        m.fc = nn.Linear(m.fc.in_features, num_classes)
        arch_name = "ResNet-50"
    elif model_name == "victorio":
        m = nn.Sequential(
            nn.Conv2d(3, 16, 3, 1, 1), nn.ReLU(), nn.MaxPool2d(2),
            nn.Conv2d(16, 32, 3, 1, 1), nn.ReLU(), nn.MaxPool2d(2),
            nn.AdaptiveAvgPool2d((4, 4)),
            nn.Flatten(),
            nn.Linear(32*4*4, num_classes)
        )
        arch_name = "Victorio CNN"
    else:
```

```

        raise ValueError(...)

    # Unified print after all branches.
    print(f'Model built: {arch_name} ({num_classes} classes, {img_size}px{
↪input})')
    return m

```

0.6 6. Evaluation Helpers.

Define helper functions to perform model evaluation and generate confusion matrix plots.

Run inference on the test dataset to compute key performance metrics such as accuracy, precision, recall, F1-score, and AUROC (multi-class One-vs-Rest). Implement a separate plotting utility to visualise confusion matrices and

export them as publication-quality figures. ##### Rationale: - Automate quantitative evaluation of all trained models under a unified procedure.

- Compute robust metrics for balanced and imbalanced classes using macro-averaging.
- Include AUROC for probabilistic assessment of multi-class separability.
- Generate reproducible visual summaries for reporting and model comparison.

```

[7]: # Evaluation Helpers (single pass, AUROC optional).
def evaluate_model(model: nn.Module, loader) -> dict:
    """Run inference over test loader; return metrics dict."""
    model.eval()
    all_logits, all_preds, all_labels = [], [], []

    with torch.no_grad():
        for imgs, lbls in loader:
            imgs = imgs.to(DEVICE)
            lbls = lbls.to(DEVICE)
            logits = model(imgs)
            preds = logits.argmax(1)

            all_logits.append(logits.cpu())
            all_preds.append(preds.cpu())
            all_labels.append(lbls.cpu())

    logits = torch.cat(all_logits, dim=0).numpy()
    preds = torch.cat(all_preds, dim=0).numpy()
    labels = torch.cat(all_labels, dim=0).numpy()

    acc = accuracy_score(labels, preds)
    prec, rec, f1, _ = precision_recall_fscore_support(
        labels, preds, average="macro", zero_division=0
    )

    # AUROC (multi-class OVR) requires probabilities.

```

```

try:
    probs = (torch.from_numpy(logits)).softmax(dim=1).numpy()
    auroc = roc_auc_score(labels, probs, multi_class="ovr")
except Exception:
    auroc = float("nan")

return {
    "accuracy": float(acc),
    "precision": float(prec),
    "recall": float(rec),
    "f1_score": float(f1),
    "auroc": float(auroc),
    "logits": logits,    # for optional downstream analysis
    "preds": preds,
    "labels": labels,
}

def plot_confusion(cm, labels, title, out_path):
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=labels)
    disp.plot(cmap="Blues", values_format=".0f")
    plt.title(title)
    plt.tight_layout()
    plt.savefig(out_path, dpi=300, bbox_inches="tight")
    plt.close()

```

0.7 7. Evaluate All Checkpoints.

Evaluate all trained model checkpoints across both ImageNet and CXR normalisation domains. Automatically detect each saved checkpoint, rebuild the corresponding architecture, and apply the correct normalisation statistics. Run inference on the fixed, deduplicated test split created in 01_data_edu.ipynb to compute accuracy, precision, recall, F1-score, and AUROC, and generate confusion matrix visualisations for each model. Finally, export aggregated metrics as JSON and CSV summaries for reporting and comparison. ##### Rationale: - Ensure consistent and reproducible evaluation of all architectures under both normalisation regimes. - Automate checkpoint discovery and metric computation for efficient large-scale comparison. - Include multiple performance indicators to capture both class balance and discriminative ability. - Generate reusable evaluation summaries and figures for quantitative analysis and publication.

```

[9]: # Evaluate all checkpoints (4 models × 2 norms) on the fixed test split.

from torch.utils.data import DataLoader

found_ckpts = sorted(CHECKPOINTS_DIR.glob("*_best.pt"))
if not found_ckpts:
    raise FileNotFoundError("No *_best.pt checkpoints found in experiments/
↳ checkpoints/")

```

```

summary = {}
class_names = [INDEX_TO_CLASS[i] for i in sorted(INDEX_TO_CLASS)]

print("Discovered checkpoints:")
for p in found_ckpts:
    print(" -", p.name)

for ckpt_path in found_ckpts:
    name = ckpt_path.stem # e.g., mobilenet_v2_imagenet_best

    # Parse model + normalisation from filename.
    parts = name.split("_")
    # Assume last token is "best" and the one before is norm suffix.
    norm_suffix = parts[-2] # "imagenet" or "cxr"
    model_name = "_".join(parts[:-2]) # "mobilenet_v2", "efficientnet_b0",
    ↪ "resnet50", "victorio"

    # Pick the correct normalisation stats
    if norm_suffix == "imagenet":
        MEAN, STD, src = IMAGENET_MEAN, IMAGENET_STD, "ImageNet"
    else:
        MEAN, STD, src = CXR_MEAN, CXR_STD, "CXR"

    # Build eval transforms for this checkpoint.
    _, eval_transforms = build_shared_transforms(
        img_size=IMG_SIZE,
        mean=MEAN,
        std=STD,
    )

    # Build test dataset/loader for this checkpoint (same images, matching
    ↪ stats)
    CHEXPERT_PATH = DATA_DIR / "chexpert"
    TBX11K_PATH = DATA_DIR / "tbx11k"

    test_dataset = CXRThreeClassDataset(
        items=test_df, # <--- use the fixed test_df here
        class_to_index=CLASS_TO_INDEX,
        transforms=eval_transforms,
        chexpert_root=CHEXPERT_PATH,
        tbx_root=TBX11K_PATH,
    )

    test_loader = DataLoader(
        test_dataset,
        batch_size=64,
        shuffle=False,

```

```

        num_workers=4,
        pin_memory=True,
    )

    print(f"\nEvaluating {name} [{src}-normalised] on {len(test_dataset)} test_
↪images...")

    # Rebuild architecture and load weights.
    model = build_model(model_name, num_classes=len(CLASS_TO_INDEX)).to(DEVICE)
    model.load_state_dict(torch.load(ckpt_path, map_location=DEVICE))

    # Run evaluation.
    metrics = evaluate_model(model, test_loader)

    # Confusion matrix.
    cm = confusion_matrix(metrics["labels"], metrics["preds"])
    fig_path = FIG_DIR / f"{name}_confusion_matrix.png"
    plot_confusion(cm, class_names, f"Confusion Matrix: {name}", fig_path)

    print(
        f' Acc={metrics["accuracy"]:.4f} '
        f' Prec={metrics["precision"]:.4f} '
        f' Rec={metrics["recall"]:.4f} '
        f' F1={metrics["f1_score"]:.4f} '
        f' AUROC={metrics["auroc"]:.4f}'
    )
    print(f" Saved: {fig_path.name}")

    # Store summary (excluding bulky arrays).
    summary[name] = {
        "model": model_name,
        "normalisation": src,
        "accuracy": metrics["accuracy"],
        "precision": metrics["precision"],
        "recall": metrics["recall"],
        "f1_score": metrics["f1_score"],
        "auroc": metrics["auroc"],
    }

    # Persist JSON + CSV summaries.
    summary_json = TABLE_DIR / "test_evaluation_summary.json"
    with open(summary_json, "w") as f:
        json.dump(summary, f, indent=2)

    summary_csv = TABLE_DIR / "test_evaluation_summary.csv"
    with open(summary_csv, "w", newline="") as f:
        writer = csv.writer(f)

```

```

writer.
writerow(["name", "model", "normalisation", "accuracy", "precision", "recall", "f1_score", "auroc"])
for name, s in summary.items():
    writer.writerow([
        name, s["model"], s["normalisation"],
        f'{s["accuracy"]:.6f}', f'{s["precision"]:.6f}',
        f'{s["recall"]:.6f}', f'{s["f1_score"]:.6f}', f'{s["auroc"]:.6f}'
    ])

print("\nSummary written to:")
print(" -", summary_json.relative_to(PROJECT_ROOT))
print(" -", summary_csv.relative_to(PROJECT_ROOT))

# Print to notebook.
print("\nFinal Test Results")
print(f'{"Checkpoint":34s} | {"Norm":9s} | {"Acc":>6s} | {"Prec":>6s} | {"Rec":>6s} | {"F1":>6s} | {"AUROC":>7s}')
print("-"*80)
for name, s in sorted(summary.items()):
    print(f'{name:34s} | {s["normalisation"][:9]:9s} | {s["accuracy"]:.3f} | {s["precision"]:.3f} | {s["recall"]:.3f} | {s["f1_score"]:.3f} | {s["auroc"]:.3f}')

```

Discovered checkpoints:

- efficientnet_b0_cxr_best.pt
- efficientnet_b0_imagenet_best.pt
- mobilenet_v2_cxr_best.pt
- mobilenet_v2_imagenet_best.pt
- resnet50_cxr_best.pt
- resnet50_imagenet_best.pt
- victorio_cxr_best.pt
- victorio_imagenet_best.pt

Evaluating efficientnet_b0_cxr_best [CXR-normalised] on 1353 test images...

Model built: EfficientNet-B0 (3 classes, 256px input)

Acc=0.9985 Prec=0.9985 Rec=0.9985 F1=0.9985 AUROC=1.0000

Saved: efficientnet_b0_cxr_best_confusion_matrix.png

Evaluating efficientnet_b0_imagenet_best [ImageNet-normalised] on 1353 test images...

Model built: EfficientNet-B0 (3 classes, 256px input)

Acc=0.9978 Prec=0.9977 Rec=0.9978 F1=0.9977 AUROC=0.9998

Saved: efficientnet_b0_imagenet_best_confusion_matrix.png

Evaluating mobilenet_v2_cxr_best [CXR-normalised] on 1353 test images...

Model built: MobileNetV2 (3 classes, 256px input)

Acc=1.0000 Prec=1.0000 Rec=1.0000 F1=1.0000 AUROC=1.0000

Saved: mobilenet_v2_cxr_best_confusion_matrix.png

Evaluating mobilenet_v2_imagenet_best [ImageNet-normalised] on 1353 test images...

Model built: MobileNetV2 (3 classes, 256px input)

Acc=0.9993 Prec=0.9992 Rec=0.9993 F1=0.9992 AUROC=1.0000

Saved: mobilenet_v2_imagenet_best_confusion_matrix.png

Evaluating resnet50_cxr_best [CXR-normalised] on 1353 test images...

Model built: ResNet-50 (3 classes, 256px input)

Acc=0.9993 Prec=0.9992 Rec=0.9993 F1=0.9992 AUROC=1.0000

Saved: resnet50_cxr_best_confusion_matrix.png

Evaluating resnet50_imagenet_best [ImageNet-normalised] on 1353 test images...

Model built: ResNet-50 (3 classes, 256px input)

Acc=0.9993 Prec=0.9992 Rec=0.9993 F1=0.9992 AUROC=0.9990

Saved: resnet50_imagenet_best_confusion_matrix.png

Evaluating victorio_cxr_best [CXR-normalised] on 1353 test images...

Model built: Victorio CNN (3 classes, 256px input)

Acc=0.9217 Prec=0.9209 Rec=0.9217 F1=0.9204 AUROC=0.9800

Saved: victorio_cxr_best_confusion_matrix.png

Evaluating victorio_imagenet_best [ImageNet-normalised] on 1353 test images...

Model built: Victorio CNN (3 classes, 256px input)

Acc=0.9084 Prec=0.9075 Rec=0.9082 F1=0.9072 AUROC=0.9779

Saved: victorio_imagenet_best_confusion_matrix.png

Summary written to:

- reports/tables/test_evaluation_summary.json
- reports/tables/test_evaluation_summary.csv

Final Test Results

Checkpoint	Norm	Acc	Prec	Rec
F1 AUROC				
efficientnet_b0_cxr_best	CXR	0.999	0.998	0.998
1.000				
efficientnet_b0_imagenet_best	ImageNet	0.998	0.998	0.998
1.000				
mobilenet_v2_cxr_best	CXR	1.000	1.000	1.000
1.000				
mobilenet_v2_imagenet_best	ImageNet	0.999	0.999	0.999
1.000				
resnet50_cxr_best	CXR	0.999	0.999	0.999
1.000				
resnet50_imagenet_best	ImageNet	0.999	0.999	0.999
0.999				
victorio_cxr_best	CXR	0.922	0.921	0.920

```
0.980
victorio_imagenet_best          | ImageNet | 0.908 | 0.907 | 0.908 | 0.907 |
0.978
```

Notes:

- **Accuracy (Acc):** Proportion of all test images correctly classified.
- **Precision (Prec):** Among all images the model predicted as belonging to a class, how many were actually correct.
- **Recall (Rec):** Among all images that really belong to a class, how many the model managed to detect.
- **F1-score (F1):** Harmonic mean of precision and recall.
- **AUROC (Area Under the Receiver Operating Characteristic Curve):** Measures how well the model separates classes across all possible decision thresholds.

0.8 8. Visual Summary of Test Accuracy.

Generate a bar chart comparing test accuracy across all evaluated models and normalisation domains.

Visualise results directly in the notebook and save the figure for inclusion in the final report.

The chart provides a clear, high-level comparison of performance between transfer learning (ImageNet)

and dataset-specific (CXR) training strategies. ##### Rationale: - Present an intuitive visual summary of quantitative results across all models.

- Highlight relative differences in performance between normalisation regimes.

- Facilitate quick inspection of accuracy trends before detailed interpretability analysis.

- Store the figure in the reports directory for reproducibility and documentation.

```
[10]: labels = list(summary.keys())
      accs = [summary[k]["accuracy"] for k in labels]

      plt.figure(figsize=(10, 4))
      bars = plt.bar(range(len(labels)), accs, color="steelblue")

      plt.xticks(range(len(labels)), labels, rotation=45, ha="right")
      plt.ylabel("Test Accuracy")
      plt.title("Test Accuracy: All Models (ImageNet vs CXR)")

      # Add white text labels inside each bar
      for bar, acc in zip(bars, accs):
          plt.text(
              bar.get_x() + bar.get_width() / 2,
              bar.get_height() - 0.02,          # slight downward offset
              f"{acc:.3f}",
              ha="center", va="top",
              color="white", fontsize=10, fontweight="bold"
          )
```

```
plt.tight_layout()

out_png = FIG_DIR / "test_accuracy_all_models.png"
plt.savefig(out_png, dpi=300, bbox_inches="tight")
plt.close()
print("Saved figure:", out_png.relative_to(PROJECT_ROOT))

display(Image(filename=FIG_DIR / "test_accuracy_all_models.png"))
```

Saved figure: reports/figures/test_accuracy_all_models.png

```
-----
NameError                                Traceback (most recent call last)
Cell In[10], line 28
     25 plt.close()
     26 print("Saved figure:", out_png.relative_to(PROJECT_ROOT))
--> 28 display(Image(filename=FIG_DIR / "test_accuracy_all_models.png"))

NameError: name 'Image' is not defined
```

0.9 9. Confusion Matrix Comparison

Display all confusion matrices in a 2×4 grid layout to compare model performance across normalisation domains.

Arrange ImageNet-based models on the top row and CXR-based models on the bottom row for direct visual comparison.

Each matrix illustrates class-specific performance, highlighting which categories were most frequently misclassified. ##### Rationale: - Provide a structured visual overview of model confusion across all architectures.

- Enable direct comparison between ImageNet-initialised and CXR-trained variants.
- Identify systematic misclassifications and potential data-driven biases.
- Support qualitative interpretation of quantitative metrics presented earlier.

```
[ ]: # Visualise All Confusion Matrices in Grid Layout
import matplotlib.pyplot as plt
from PIL import Image as PILImage

# Collect all confusion matrix images from the reports/figures folder.
confusion_paths = sorted(FIG_DIR.glob("*_confusion_matrix.png"))
if not confusion_paths:
    raise FileNotFoundError("No confusion matrix PNG files found in reports/
    figures.")

# Separate into ImageNet and CXR variants.
imagenet_paths = sorted([p for p in confusion_paths if "imagenet" in p.stem])
cxr_paths       = sorted([p for p in confusion_paths if "cxr" in p.stem])
```

```

# Sanity check: both should contain the same 4 model names.
if len(imagenet_paths) != len(cxr_paths):
    print(f'Warning: Unequal number of ImageNet ({len(imagenet_paths)}) and CXR_
    ↪({len(cxr_paths)}) matrices detected.')

# Create grid: 2 rows (normalisation) × 4 columns (architectures).
fig, axes = plt.subplots(2, len(imagenet_paths), figsize=(16, 8))
fig.suptitle("Confusion Matrices: ImageNet vs CXR Normalisation", fontsize=14,
    ↪weight="bold")

# Top Row: ImageNet.
for j, path in enumerate(imagenet_paths):
    img = PILImage.open(path)
    axes[0, j].imshow(img)
    model_name = path.stem.replace("_imagenet_best_confusion_matrix", "")
    axes[0, j].set_title(f'{model_name} (ImageNet)', fontsize=10)
    axes[0, j].axis("off")

# Bottom Row: CXR.
for j, path in enumerate(cxr_paths):
    img = PILImage.open(path)
    axes[1, j].imshow(img)
    model_name = path.stem.replace("_cxr_best_confusion_matrix", "")
    axes[1, j].set_title(f'{model_name} (CXR)', fontsize=10)
    axes[1, j].axis("off")

# Adjust spacing and show final grid.
plt.tight_layout(rect=[0, 0, 1, 0.95])

# Save BEFORE showing. This avoids clearing the figure.
out_path = FIG_DIR / "confusion_matrix_grid_2x4.png"
plt.savefig(out_path, dpi=300, bbox_inches="tight")
print(f"Composite grid saved to: {out_path.relative_to(PROJECT_ROOT)}")

plt.show()

```

```
[ ]:
```